

AI PROBLEM SCOPING

AI Placement Preparation Assistant



[Figure 1.0: Predictive Placement Optimization & Student Deficit Funnel]

A structured 8-step scoping framework designed for translating a university recruitment request into a measurable, interactive, and predictive machine-learning solution.

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STEP AGENDA

Eight steps from university observation to AI deployment solution:

- **01** Understanding the Business Problem
- **02** Defining the Problem Precisely
- **03** Is AI Really Needed?
- **04** Identifying Stakeholders
- **05** Defining the AI Task
- **06** Defining the Success Metrics
- **07** Defining Constraints
- **08** Assumptions and Risks

01. UNDERSTANDING THE BUSINESS PROBLEM

Initial Statement: University management notices that many students are rejected during the campus interview. Management wants to increase the placement ratio. They approach an AI team/organization and ask to build an AI assistant for placement preparation and to find the weakness points of the students.

Stakeholder Discovery Matrix:

Academic & Placement Cell Signals

Current placement ratio dropped to 20% against a target of 30% since 2024. Average student CGPA stands at 7.5 while major corporations mandate a strict eligibility baseline of 8.0 CGPA.

Student Feedback Matrix

Students dedicate 2 hours per day over 3 years for interview tracks. Primary identified failure bottlenecks include verbal communication, advanced data structures, and structural engineering mock round handling.



[Figure 2.0: Student Competency Matrix & Interview Deficit Analytics Dashboard]

02. DEFINING THE PROBLEM PRECISELY

POOR PROBLEM STATEMENT

"Prepare students for placement"

BETTER APPROACH (PRECISE STATEMENT)

"Help 2nd year student start preparation for placement and it is completed around 3rd year so that they can improve their chances of getting placed in their desired company"

The Precision Mapping (5 Ws Framework)

Who?	What?	When?	Why?	Outcome?
2nd year student	Preparation for interview	Starting from 2nd year and continue until end of 3rd year	So that you achieve your desired company	Increase the placement ratio

03. IS AI REALLY NEEDED?

Core Scoping Question: Can the problem be solved without AI?

Non-AI Benchmark: Student chooses any online batch, coaching center for preparation for placement.

Why Machine Learning (ML)?

AI should only be used when patterns are complex, multiple factors interact, and manual rules are insufficient. In this case, placement is dependent on many dynamic factors:

- Communication skill
- CGPA
- Aptitude skills
- Problem solving skill
- How to handle the things
- Projects
- Behavioural skill
- Interview performance

Conclusion: Patterns are going to be difficult → AI is needed

04. IDENTIFYING STAKEHOLDERS

Stakeholder Mapping Matrix:

- **University Management:** Increase placements ratio
- **Students:** Prepare for interview
- **Parents:** Student success
- **Placement Officer:** Help more students get placed
- **Faculty:** Improve student outcome
- **Placement Cell:** Organize preparation activities

05. DEFINING THE AI TASK

Translating objectives into explicit mathematical/computational targets:

- **Placement Prediction:** Classification problem. Output → **Placed = Yes / Placed = No** (Binary classification problem)
- **Ranking Problem:** Top student getting placed first
- **Identify Weak Area of Each Student:** Students work on and perform better in interview
- **Suggest Companies Match with Student Profile:** Student give interview in that companies and getting placed

06. DEFINING THE SUCCESS METRICS

Technical Metrics

To evaluate algorithmic execution:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Business Metrics

To measure real-world campus impact:

- Placement ratio improve
- More student get placed
- Aptitude score improve
- Mock interview score improve

07. DEFINING CONSTRAINTS

Every AI project operates under explicit real-world constraints:

- **Time Constraints:** Process must complete before end of the 3rd year.
- **Data Constraints:** Certain information may not exist, creating blind spots:
 - Mental health indicator
 - Family circumstances
 - Updated financial records
- **Ethical Constraints:**
 - Student should not be harmed.
 - Is the model biased against a particular group?
 - Can student appeal decisions?
 - Student data is not shared to any other company without permission.

08. ASSUMPTIONS AND RISKS

Assumptions

- Historical data reflects future trend.
- CGPA records are accurate.

Risks

- Missing data
- Biased historical decisions
- Unexpected events that affect placement ratio
- Changes in academic policies
- Internet failure

Thank you.

Questions on placement scoping, algorithmic choices, or weak-point mapping frameworks — happy to discuss.